

Embedded Systems & IoT Internship – Task 2

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Role

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Department

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Title

Temperature Detection System Using TMP36 and Arduino UNO

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Introduction

Temperature monitoring systems are widely used in homes, industries, and environmental monitoring applications. This project is designed using Arduino UNO and a TMP36 temperature sensor to measure surrounding temperature in real time.

The TMP36 sensor detects temperature changes and sends the data to the Arduino. The Arduino processes the sensor values and displays the temperature in the Serial Monitor. When the temperature exceeds the predefined threshold value, the LED automatically turns ON as an alert indication.

This project helps in understanding sensor interfacing, embedded systems, and basic automation using Arduino and Tinkercad simulation.

Objective

- To interface the TMP36 temperature sensor with Arduino UNO.
- To monitor temperature using sensor data.
- To display temperature readings in Serial Monitor.
- To trigger an LED alert when temperature exceeds the threshold value.
- To understand embedded systems and sensor interfacing concepts.

Components Required

- Arduino UNO
- Breadboard
- TMP36 Temperature Sensor
- LED

- 220Ω Resistor
- Jumper Wires
- Tinkercad Simulation Platform

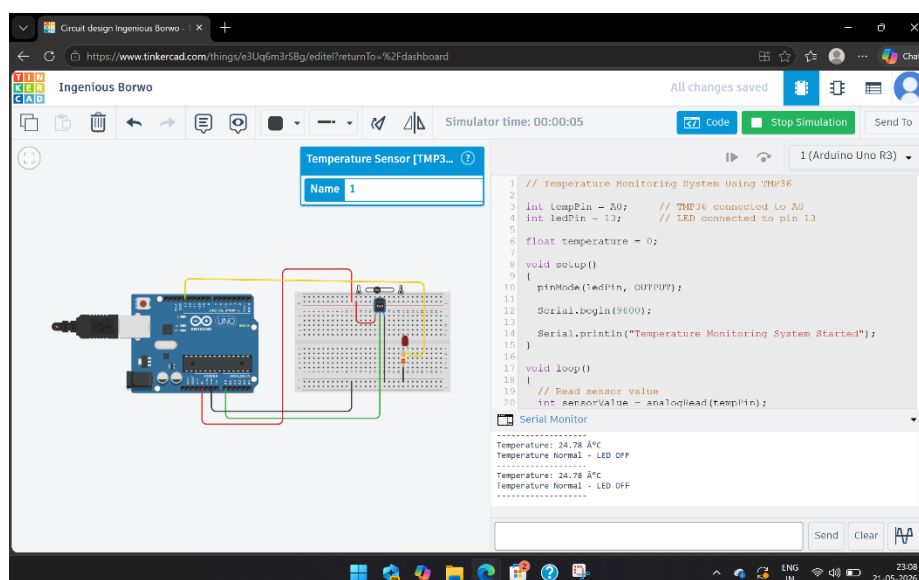
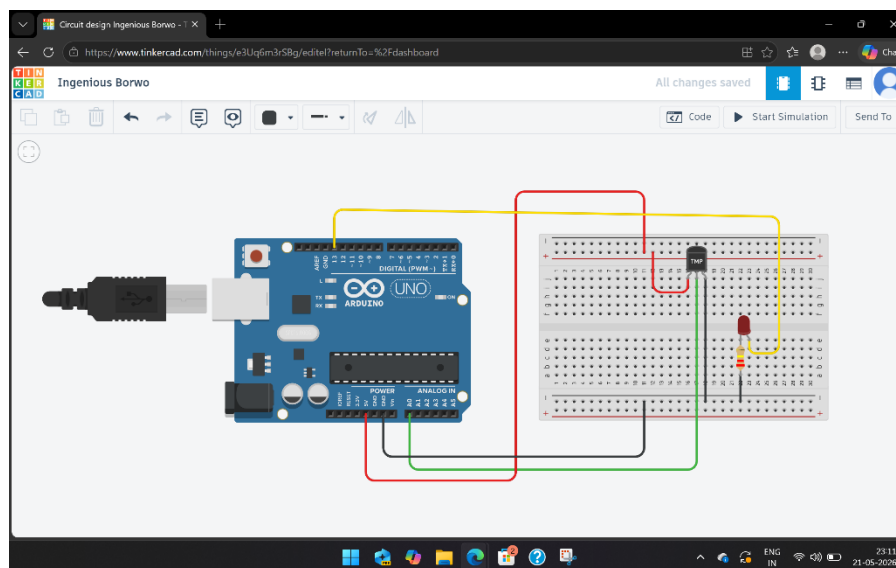
Working Principle

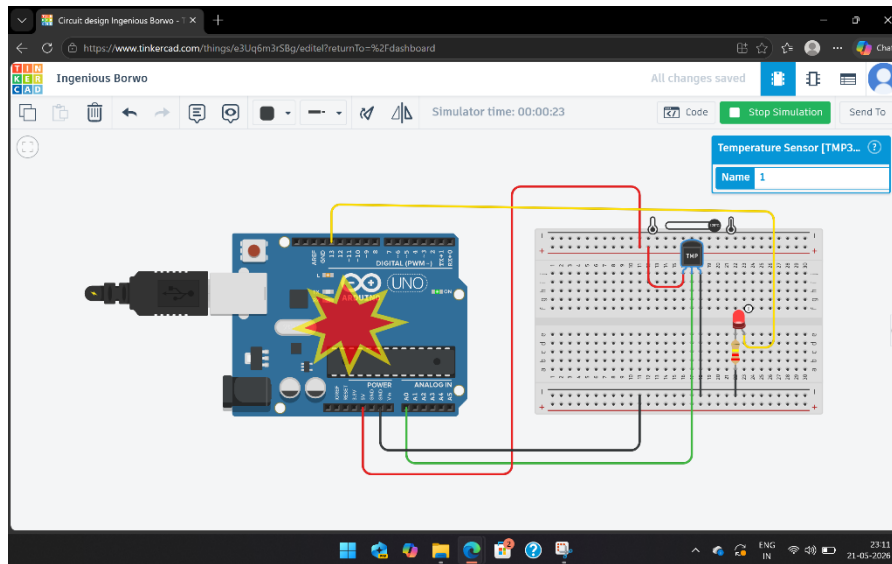
The TMP36 temperature sensor continuously senses the surrounding temperature and produces an analog voltage proportional to the temperature value. Arduino UNO reads this analog signal through pin A0.

The analog value is converted into voltage and then into temperature in degree Celsius using Arduino code. The temperature readings are displayed in the Serial Monitor.

A threshold value of 30°C is used in the project. When the measured temperature exceeds 30°C, Arduino automatically turns ON the LED to indicate high temperature. When the temperature falls below 30°C, the LED turns OFF.

Circuit Diagram





Arduino Code

// Temperature Monitoring System Using TMP36

int tempPin = A0; // TMP36 connected to A0

int ledPin = 13; // LED connected to pin 13

float temperature = 0;

void setup()

{

pinMode(ledPin, OUTPUT);

Serial.begin(9600);

Serial.println("Temperature Monitoring System Started");

}

void loop()

{

// Read sensor value

int sensorValue = analogRead(tempPin);

// Convert analog value to voltage

float voltage = sensorValue * (5.0 / 1023.0);

// Convert voltage to temperature in Celsius

temperature = (voltage - 0.5) * 100;

// Display temperature

Serial.print("Temperature: ");

Serial.print(temperature);

```

Serial.println(" °C");
// Threshold condition
if (temperature > 30)
{
    digitalWrite(ledPin, HIGH);
    Serial.println("High Temperature Alert - LED ON");
}
else
{
    digitalWrite(ledPin, LOW);
    Serial.println("Temperature Normal - LED OFF");
}
Serial.println("-----");
delay(1000);
}

```

Code Explanation

The Arduino program is used to read temperature values from the TMP36 temperature sensor and control an LED based on the temperature condition.

Output

Condition 1: Temperature Below 30°C

- LED remains OFF.
- Serial Monitor displays normal temperature readings.

Condition 2: Temperature Above 30°C

- LED turns ON.
- Serial Monitor displays “High Temperature Alert - LED ON”.

Conclusion

- The Smart Temperature Monitoring System using TMP36 and Arduino UNO was successfully implemented using the Tinkercad simulation platform. The system continuously monitored temperature values and activated an LED alert when the temperature exceeded the threshold value.
- This project helped in understanding sensor interfacing, embedded C/C++ programming, Arduino UNO operation, and real-time monitoring systems.